Research Report

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Introduction

Electric vehicle (EV) prices vary substantially across models, even among vehicles with similar technical specifications. Battery capacity is one of the most visible EV characteristics and is often associated with both production cost and consumer value. However, vehicle price is not determined by battery capacity alone. Manufacturers differ in brand positioning, and premium brands may charge systematically higher prices than non-premium brands even for vehicles with similar specifications. Understanding this relationship is important for both consumers and manufacturers. For consumers, it helps clarify whether higher prices reflect meaningful technical improvements or brand positioning. For manufacturers, it provides insight into how battery capacity is translated into pricing across different market segments.

This report studies EV models listed for sale in Germany and investigates whether battery capacity is associated with vehicle price, and whether that association differs between premium and non-premium manufacturers. The research questions are:

1. Among electric vehicle models listed for sale in Germany, is battery capacity positively associated with vehicle price?
2. Does the association between battery capacity and price differ between premium and non-premium manufacturers?

The data come from the *Electric Vehicle Specifications and Prices* dataset obtained from Kaggle and compiled by web scraping EV Database. EV Database aggregates manufacturer-reported technical specifications and listed prices for electric vehicle models listed for sale in Germany. The Kaggle dataset page indicates that the dataset entry was updated "2 years ago," which provides some background on when the posted dataset was last refreshed. However, that platform update note should not be interpreted as a precise statement of the model year coverage or the exact scrape date of the underlying EV listings. The dataset contains 360 observations, where each row represents a distinct EV model. This dataset should not be interpreted as a complete census of all electric vehicles that have ever existed or all EVs ever sold in Germany. Instead, it is best understood as a snapshot of the EV models included in EV Database when the source was scraped and the Kaggle dataset was assembled. The exact calendar year or scrape date is not stated in the raw CSV used here, so the report avoids treating the data as representing a single fully specified year. Since the data are observational rather than experimental, all conclusions in this report are associative rather than causal.

The response variable is `Price.DE.`, the listed vehicle price in euros. The main explanatory variable is `Battery`, measured in kilowatt-hours (kWh). A second key variable is manufacturer segment, coded as premium or non-premium. Following the proposal, Tesla, BMW, Audi, Mercedes-Benz, and Porsche are classified as premium brands based on their widely recognized positioning in the European automotive market. This grouping is intended to reflect broad brand-tier differences rather than a formal regulatory classification. Other quantitative variables that may be useful during model building include `Range`, `Efficiency`, `Top_speed`, `Acceleration..0.100.`, and `Fast_charge`. The exploratory analysis is conducted on the original variable scales. For the main regression analysis, a log transformation of price is later adopted after diagnostics indicate that the untransformed price model does not adequately satisfy the linear-model assumptions. The main structural modification to the raw data is the premium versus non-premium grouping at the manufacturer level.

This report uses multiple linear regression because the response is continuous and the main goal is inference about average relationships. In particular, including an interaction between battery capacity and market segment allows the battery-price relationship to differ between premium and non-premium manufacturers, which directly addresses the second research question. Based on the course material, variables named in the research question remain in the model throughout model selection even if a selection procedure would otherwise remove them.

Exploratory Analysis

Data Preparation

The raw dataset is stored locally in the project folder as `data/EV_cars.csv`. Using a relative file path makes the analysis reproducible for all group members and avoids machine-specific paths such as `~/Desktop/...`.

The raw file contains 360 observations and 9 recorded variables: `Battery`, `Car_name`, `Car_name_link`, `Efficiency`, `Fast_charge`, `Price.DE.`, `Range`, `Top_speed`, and `acceleration..0.100.`. Because the data were scraped from an online database, the observed models reflect what EV Database included during the underlying collection period, not necessarily every EV available in the German market before or after that time. This matters for interpretation: the analysis is best understood as describing relationships among the EV models represented in this snapshot, rather than all EVs from a specific stated year. The response variable `Price.DE.` has 51 missing values, and `Fast_charge` has 2 missing values. Since price is the response, rows with missing price cannot be used for the regression analysis and are removed. The two rows with missing `Fast_charge` are retained during initial cleaning and are only excluded in models that include `Fast_charge`.

To prepare the analysis dataset, a manufacturer variable was extracted from the vehicle name, the premium/non-premium grouping from the proposal was applied, the acceleration variable was renamed to a cleaner label, and only the variables relevant to the analysis were retained. Identifier variables such as `Car_name_link` were not used as predictors because they do not have an interpretable statistical relationship with price. Manufacturer names are extracted heuristically and may introduce minor misclassification for multi-word brands.

After removing rows with missing price, the main analysis dataset contains 309 EV models. A check of duplicate `Car_name` values shows that some names appear more than once. These rows were retained because they are not exact duplicates of the full record. In several cases, the repeated names correspond to different EV Database links and slightly different technical specifications, which suggests updated entries or distinct configurations rather than accidental duplication. For example, duplicated Mercedes entries often appear as an older version with missing price and a newer version with a reported price, while the duplicated Mini Cooper SE entries differ substantially in battery capacity and range and therefore represent genuinely different configurations.

For the main regression, one practical choice is to fit models using complete cases on the variables included in each model. If `Fast_charge` remains in a candidate model, the effective sample size becomes 307 rather than 309 because of the two missing charging values. This should be noted when comparing models. To assess whether the 51 models with missing prices appeared systematically different from the retained models, the observed technical variables were compared across the two groups during data preparation. Those comparisons did not suggest large differences in battery capacity, range, efficiency, top speed, acceleration, or fast-charge capability, which reduces concern that the missing prices are concentrated only among very low-end or very high-end vehicles. However, because the missingness mechanism is not known, some selection bias remains possible.

Exploratory Summaries

Exploratory analysis serves two purposes in this study. First, it helps describe the data to readers who are not familiar with the EV market. Second, it helps identify features of the data that affect the regression analysis, such as skewness, segment differences, and multicollinearity among technical variables.

Start with summary statistics (Table 1) for the response and candidate explanatory variables.

Table 1: Summary statistics for key quantitative variables

N	Mean price	Median price	SD price	Mean battery	Median battery	SD battery	Mean range	Mean efficiency
309	67264.28	56942	34508.25	71.08	70.5	20.65	369.13	195.41

The 309 EV models used in the main analysis have a mean listed price of 67,264 EUR and a median price of 56,942 EUR, suggesting some right-skewness caused by a smaller number of high-priced vehicles (Table 1). Battery capacity is more balanced, with a mean of 71.1 kWh and a median of 70.5 kWh. Overall, the summary statistics indicate that price is more variable than battery capacity. Additional grouped summaries, not shown in Table 1, indicate that premium vehicles tend to have both larger batteries and higher listed prices on average than non-premium vehicles. These patterns motivate examining the joint relationship between price, battery capacity, and market segment in the regression analysis.

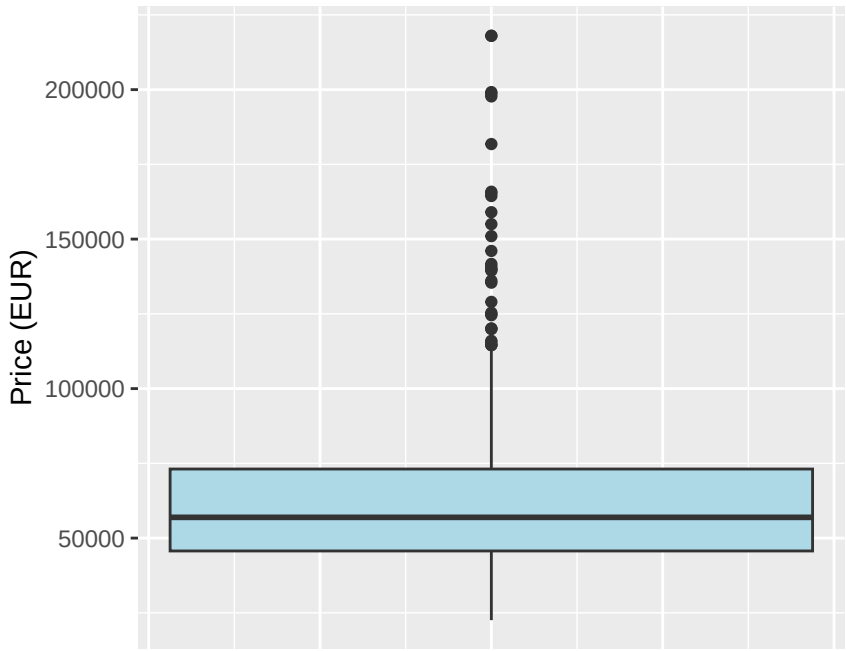

Figure 1: Boxplot of EV prices (EUR)

The boxplot of price further highlights its distributional features (Figure 1). In particular, the presence of a long upper tail and several high-price outliers confirms that price is right-skewed. Consistent with this visual evidence, the summary statistics suggest two important features that matter for modeling: price is more variable and somewhat right-skewed, while premium vehicles tend to have both larger batteries and higher prices. This motivates focusing on the joint relationship between price, battery capacity, and market segment.

The main exploratory visualization plots price against battery capacity with separate fitted lines for premium and non-premium manufacturers (Figure 2). This figure directly addresses both research questions by showing both the overall battery-price association and the extent to which that association may differ by market segment.

Separate simple linear regressions of price on battery capacity were fitted within each market segment, and the resulting slope estimates are reported in Table 2. For non-premium vehicles, the estimated increase in price is 896.68 EUR per additional kWh of battery capacity, with a standard error of 57.47. For premium

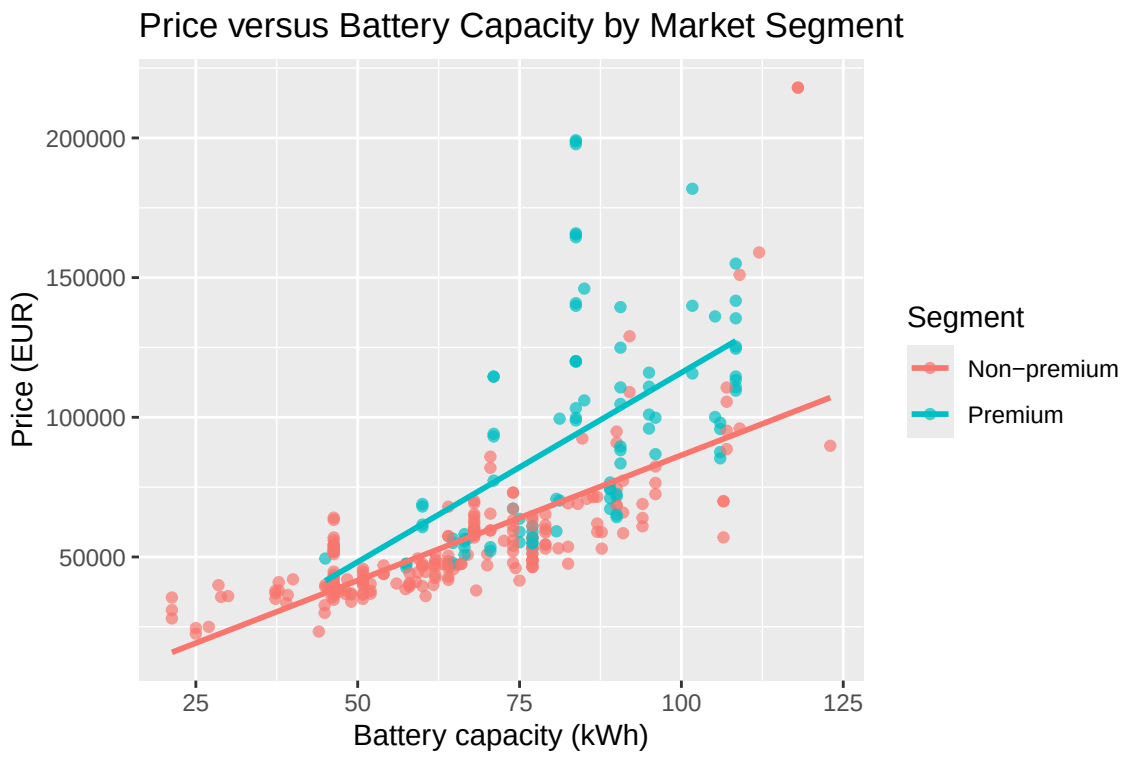

Figure 2: Price versus battery capacity by market segment

vehicles, the corresponding estimate is substantially larger at 1,355.14 EUR per kWh, with a standard error of 222.63. In both segments, the p-values are smaller than 0.001, indicating strong statistical evidence of a positive association between battery capacity and price. Consistent with the visual pattern in Figure 2, these estimates confirm that while the relationship is positive for both segments, it is noticeably steeper among premium manufacturers. This suggests that, on average, an additional kWh of battery capacity is associated with a larger increase in listed price in the premium segment.

Table 2: Estimated battery-price relationship by market segment

Segment	Slope estimate (EUR/kWh)	Standard error	p-value
Non-premium	896.68	57.47	<0.001
Premium	1355.14	222.63	<0.001

Because several quantitative vehicle characteristics may be related to one another, it is also useful to examine pairwise relationships and correlations. This can help identify potential multicollinearity before model fitting.

Table 3: Correlation matrix for quantitative variables

	Price.DE.	Battery	Range	Efficiency	Top_speed	Acceleration_0_100	Fast_charge
Price.DE.	1.00	0.70	0.59	0.17	0.76	-0.53	0.62
Battery	0.70	1.00	0.88	0.16	0.70	-0.61	0.59
Range	0.59	0.88	1.00	-0.31	0.76	-0.75	0.72
Efficiency	0.17	0.16	-0.31	1.00	-0.21	0.42	-0.29
Top_speed	0.76	0.70	0.76	-0.21	1.00	-0.84	0.79
Acceleration_0_100	-0.53	-0.61	-0.75	0.42	-0.84	1.00	-0.68
Fast_charge	0.62	0.59	0.72	-0.29	0.79	-0.68	1.00

Potential multicollinearity can be assessed from the correlation matrix of the main quantitative vehicle characteristics (Table 3). Several strong pairwise relationships are evident. Price is strongly positively correlated with Battery (0.70) and Top_speed (0.76), indicating that more expensive vehicles tend to have larger batteries and higher performance. At the same time, Battery and Range are themselves highly correlated (0.88), suggesting that these variables capture similar underlying information about vehicle capability. In addition, Top_speed and Acceleration_0_100 are strongly negatively correlated (-0.84), reflecting the fact that faster vehicles tend to have lower acceleration times. Overall, these correlations reveal substantial dependence among the technical variables. This suggests that including many of these predictors simultaneously may introduce multicollinearity, which can make regression coefficients unstable and harder to interpret. The broader pairwise relationships among the quantitative variables are shown in Appendix Figure 5.

Statistical Analysis

Modeling Strategy

The study employs a multiple linear regression framework because the response variable is continuous and the goal is inference about average relationships. The model is built around the research-question terms: battery capacity ($Battery$), market segment ($market_segment$), and their interaction. These terms are retained throughout model selection so that both research questions remain directly addressable. Additional technical variables, including $Range$, $Efficiency$, Top_speed, $Acceleration_0_100$, and $Fast_charge$, are treated as optional controls.

Backward elimination based on partial F-tests is used for the optional controls only. This test-based approach is appropriate here because the project focuses on inference rather than prediction. Starting from the full model allows each optional variable to be assessed while the others are present, and it helps reduce the model to a more interpretable form without removing the terms required by the research questions.

Initial Model

In the raw price model, the adjusted R^2 is approximately 0.7313. However, we must verify if the assumptions of linear regression hold before trusting these results.

Diagnostic Discovery

A systematic review of the initial raw-price model's diagnostics revealed critical violations of Gauss-Markov assumptions:

- Heteroscedasticity: The "Residuals vs Fitted" plot displayed a prominent fan-shaped pattern, indicating that the error variance increases as the predicted price rises.
- Non-Normality: The Normal Q-Q plot exhibited a heavy upper tail, suggesting the residuals are right-skewed—a common trait in monetary datasets where high-end outliers are present.

Because these violations could lead to unstable coefficient estimates and unreliable p-values, we transition to a log-linear framework to stabilize the variance.

Model Refinement: Log-Transformation

To address the heteroscedasticity, we applied a natural log transformation ($log(Price)$). The backward-elimination procedure was then repeated using the log-transformed response.

Main Model Results (log linear)

The final model was fitted on 307 observations, as 2 additional rows were excluded due to missing values in `Fast_charge`. Final coefficient estimates are reported in Table 4. After backward elimination,

`Acceleration_0_100` was removed, while `Range`, `Efficiency`, `Top_speed`, and `Fast_charge` were retained as control variables. The adjusted R^2 for the final log-price model is 0.8412.

In this adjusted model, the coefficient for `Battery` in the baseline non-premium segment is not statistically significant ($p = 0.1517$). This is plausible given the strong overlap between `Battery` and other retained technical variables, especially `Range` ($r = 0.88$). In practical terms, once `Range` is already in the model, the battery coefficient asks whether two vehicles with the same range but different battery capacities still differ in price, which is a much narrower question than the simple battery-price relationship. As a result, the model should be interpreted as assessing the association of battery capacity with price after holding constant several related measures of vehicle performance.

Table 4: Coefficient estimates for the final regression model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	8.0131	0.2184	36.6899	0.0000
Battery	-0.0047	0.0033	-1.4371	0.1517
market_segmentPremium	-0.3002	0.1120	-2.6802	0.0078
Range	0.0018	0.0006	2.8726	0.0044
Efficiency	0.0075	0.0010	7.4841	0.0000
Top_speed	0.0057	0.0005	10.5893	0.0000
Fast_charge	0.0003	0.0001	3.7778	0.0002
Battery:market_segmentPremium	0.0051	0.0013	3.7781	0.0002

Interpretation of Results

Interaction Between Battery Capacity and Segment The most important result is the highly significant interaction term, $Battery : market_segmentPremium$ ($\beta = 0.0051, p < 0.001$). This coefficient represents the difference in the battery-price slope between premium and non-premium vehicles in the adjusted log-linear model. Its positive sign indicates that, after controlling for the other technical variables in the model, the association between battery capacity and price is more positive in the premium segment than in the non-premium segment. Even though the adjusted segment-specific slopes are small after conditioning on highly correlated performance variables, the interaction provides strong statistical evidence that the battery-price relationship is not the same across the two segments.

Segment-Specific Interpretation For the baseline non-premium segment, the coefficient for battery capacity is small, negative, and not statistically significant ($\beta = -0.0047, p = 0.1517$). This suggests that once range, efficiency, top speed, and charging performance are held fixed, battery capacity by itself does not show a clear independent association with listed price among non-premium vehicles.

For the premium segment, the battery slope is the sum of the baseline term and the interaction term, giving an estimated slope of $-0.0047 + 0.0051 = 0.0004$. This adjusted premium slope is therefore close to zero, but it remains higher than the corresponding non-premium slope because of the positive interaction. The key takeaway is therefore not that the adjusted battery effect is large within either segment on its own, but that the model still detects a statistically meaningful difference between segments once the other technical variables are held fixed. Taken together with the stronger unadjusted slopes shown earlier in Table 2, the fitted model suggests that much of the raw battery-price association is shared with other performance variables, while the remaining adjusted battery relationship still differs by market segment.

Control Variables The retained control variables also help explain the pricing patterns. `Range` ($p = 0.0044$), `Top_speed` ($p < 2 \times 10^{-16}$), and `Efficiency` ($p < 0.001$) each contribute information beyond battery capacity alone, indicating that the market values a broader set of performance characteristics. Their inclusion also helps explain why the adjusted coefficient for battery capacity is much smaller than the simple within-segment slopes reported in Table 2.

Diagnostics and Model Selection

Initial diagnostics of the untransformed price model revealed meaningful departures from the linear-model assumptions. As shown in Appendix Figure 3, the residual-versus-fitted plot displayed increasing spread as fitted values increased, and the Q-Q plot showed a heavy upper tail. These patterns are consistent with heteroscedasticity and right-skewness in price.

The log transformation of price reduced these problems and improved model fit, with adjusted R^2 increasing from 0.7313 to 0.8412. Some mild upper-tail deviation remains in the Q-Q plot after transformation, but it is not severe enough to overturn the main inferential conclusions. For that reason, the log-linear model is used as the primary inferential model in the report.

Discussion and Conclusion

Discussion

Taken together, the exploratory and regression results suggest that battery capacity is associated with EV price, but that relationship depends on how price is modeled and which other technical variables are held fixed. In the simple within-segment regressions, battery capacity has a clear positive association with price in both market segments, and the slope is steeper for premium vehicles. In the adjusted log-linear model, however, much of that simple association is absorbed by other correlated performance measures such as range and top speed. Under that adjusted specification, the battery coefficient is not statistically significant for non-premium vehicles, while the positive interaction term indicates that the battery-price relationship remains more favorable in the premium segment than in the non-premium segment.

This pattern is consistent with the idea that EV prices reflect a bundle of related technical characteristics rather than battery capacity alone. For non-premium vehicles in particular, battery size may matter mainly through the performance outcomes it helps produce, such as longer range or higher speed, rather than as an independently priced feature. For premium vehicles, the positive interaction suggests that battery capacity is more closely aligned with pricing even after controlling for those related characteristics, although the adjusted premium slope itself is small in the final model.

Limitations

Several limitations should be kept in mind. First, the dataset is an observational snapshot of EV models listed in EV Database and does not represent all EVs sold in Germany or all EVs from a clearly specified year. Second, the premium versus non-premium classification is a simplified grouping based on widely recognized market positioning rather than a formal industry standard. Third, some relevant determinants of price, such as interior quality, software features, or driver-assistance packages, are not recorded in the dataset. Finally, because 51 observations had missing prices and were excluded, some risk of selection bias remains even though the missing-price models do not appear dramatically different on the observed technical variables.

Conclusion

The analysis provides evidence that the association between battery capacity and EV price differs by market segment. Premium vehicles exhibit a steeper unadjusted battery-price relationship, and the adjusted log-linear model also indicates a significant difference between premium and non-premium segments through the interaction term. At the same time, the final model shows that battery capacity alone does not have a clear independent association with price for non-premium vehicles once other related technical variables are held constant. Overall, the results suggest that EV pricing reflects both battery capacity and broader performance characteristics, and that these relationships are not the same across manufacturer segments.

Statement of AI Use

Generative AI was used to help organize the structure and wording of this report and to assist in preparing a separate reproducible analysis script. The statistical conclusions, numerical results, and model interpretations are based on outputs generated from the dataset itself.

Representative prompts used:

1. "Help revise the writing so the report reads like a formal statistical report rather than notebook-style notes."
2. "Help update the data-preparation description so it accurately reflects the raw dataset, missing values, and derived variables used in the analysis."
3. "Help strengthen the wording in response to feedback on interaction interpretation, diagnostics, model selection, and limitations."
4. "Help move less essential material to the appendix so the main body remains concise and within the page limit."

Appendix

Diagnostic Plots

Supporting diagnostic plots are provided in Appendix Figure 3 and Appendix Figure 4. In Appendix Figure 4, the dashed red reference line marks the common heuristic threshold of $4/n$ for Cook's distance.

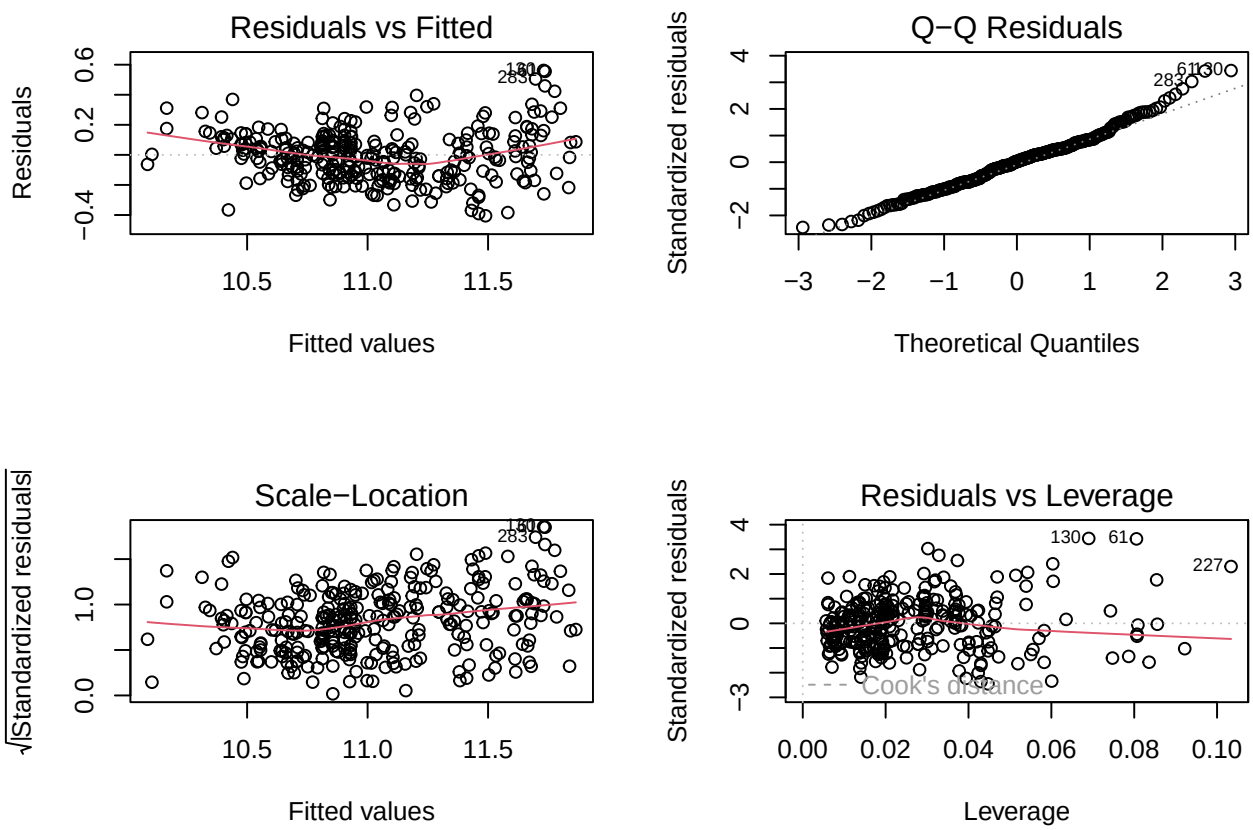

Figure 3: Diagnostic plots for the selected regression model

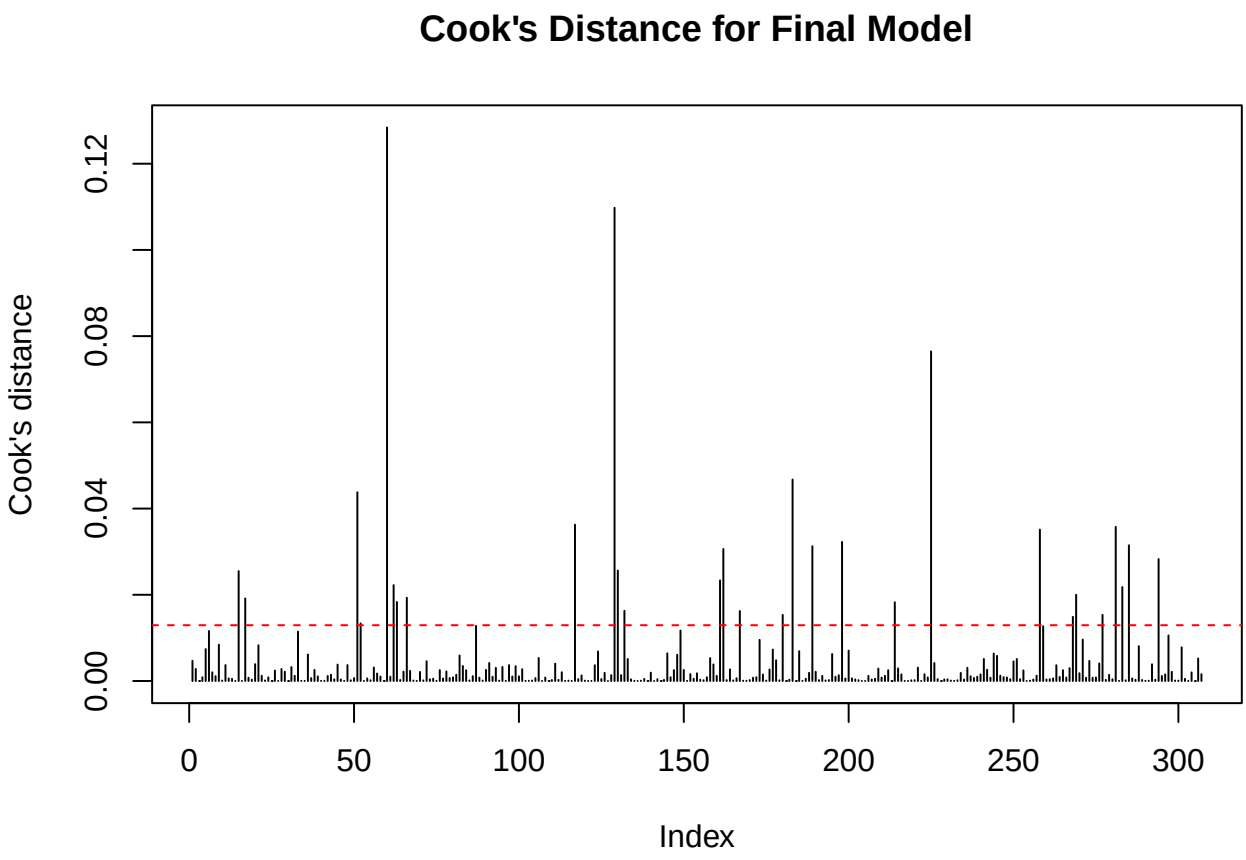

Figure 4: Cook's distance for the selected regression model

Pairwise Relationships Among Quantitative Variables

An additional exploratory check of pairwise relationships among the quantitative variables is shown in Appendix Figure 5. This figure is included in the appendix rather than the main body because it is useful for supporting model-building decisions, but it is not the most direct figure for answering the research questions.

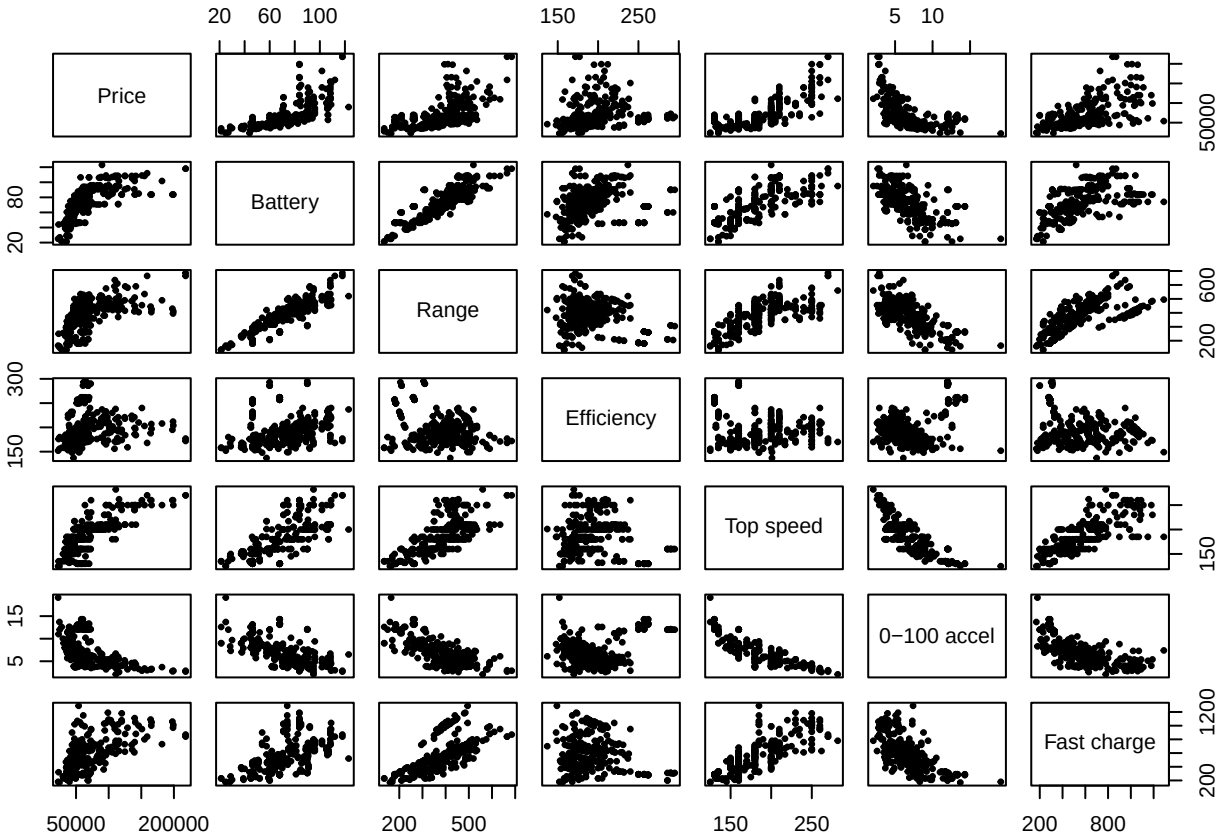

Figure 5: Scatterplot matrix for quantitative variables used in the analysis

The scatterplot matrix in Appendix Figure 5 is mainly helpful for visually checking which explanatory variables appear strongly associated with price and which predictors appear highly related to one another. In this dataset, it supports the conclusion that variables such as `Battery`, `Range`, `Top_speed`, and `Fast_charge` share substantial overlap, which helps explain why a simpler final model is preferred.

Code and data submission: Qianxue Wang